

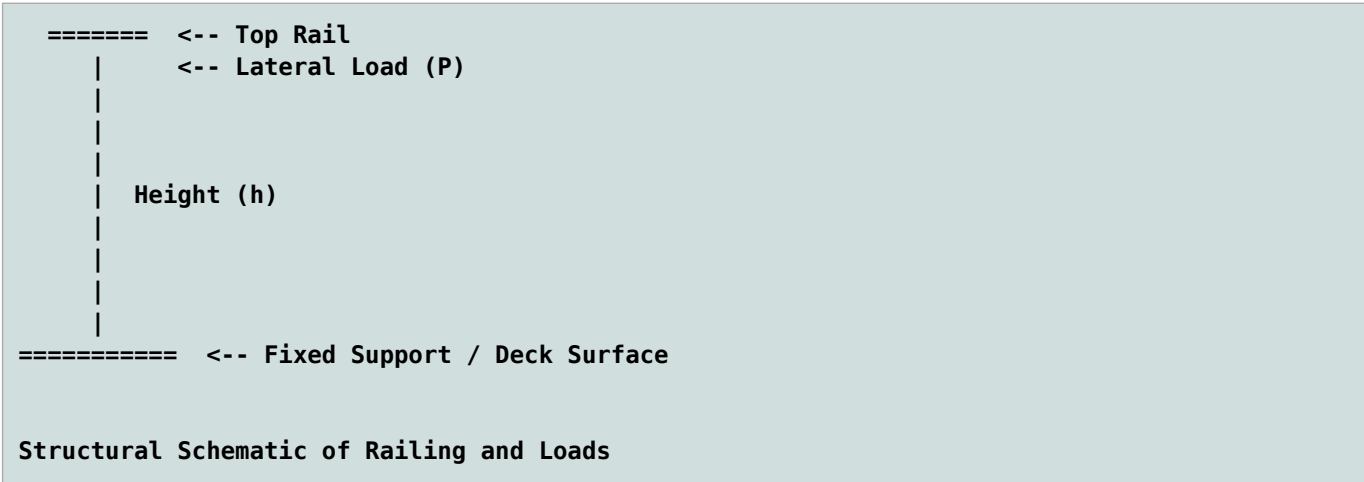
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2.2.1 Introduction

Under the California Building Code (CBC), handrails and guards (railings) must resist a uniform load of 50 plf and a concentrated point load of 200 lbs, both applied horizontally to the top rail. Intermediate rails, balusters, and infill panels must separately withstand a concentrated load of 50 lbs.



2.2.2 Railing Design

Table 1: Import Functions (rvsrc/scripts/sectprop2.py)

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL
bending_stress_udl(ln, w, b, d)	Maximum bending stress in a simply supported beam with UDL.
nds_beam_check(** kwargs)	Check stress and deflection for a simply supported wood beam using NDS.
nds_post_check(** kwargs)	Check stress at cantilever post

Design properties.

```
Function Arguments Dictionary : post1 (units: inch, pounds)
=====
h_1 = 42. # post height
P_1 = 200 # concentrated load
b_1 = 3.5 # post width
d_1 = 3.5 # post depth
E_1 = 1.5*(10**6) # modulus of elasticity
F_b = 1000 # allowable bending stress
C_D = 1.0 # load duration factor
C_M = 0.85 # wet service factor
C_F = 1.0 # size factor
C_t = 1.0 # temperature factor
C_i = 0.8 # incising factor
C_r = 1.0 # repetitive member factor
```

```
C_c = 1.0    # curvature factor
C_L = 1.0    # beam stability factor
C_b = 1.0    # bearing area factor
deflect_limit = 360.0 # max allowable deflection ln_1/deflect_limit
=====
```

Design Results

Eq. 1: Check Deck Beam

```
nds_post_check | units: inch, pounds
```

Post Check Results:

```
=====
```

```
P: 200.00
```

```
fb: 1175.51
```

```
Fb_prime: 680.00
```

```
E_prime: 1275000.00
```

```
stress_ratio: Not OK
```

```
deflection: 1.55
```

```
deflection_ratio: Not OK
```

